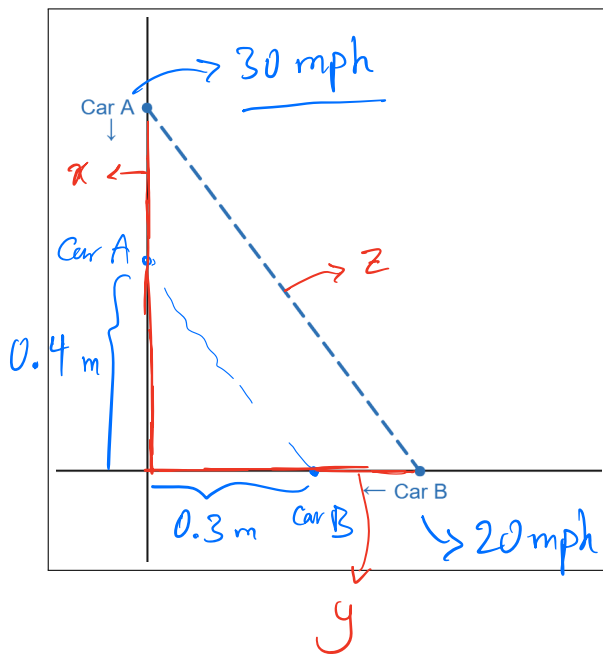


1. (Spring 2024, #7) Two cars are approaching an intersection. One car is approaching from the north and is traveling at 30 mph. The other car is approaching from the east and is traveling at 20 mph. At the moment when the north car is 0.4 miles from the intersection and the east car is 0.3 miles from the intersection, find the rate at which the distance between the two cars is changing.



$$\frac{dx}{dt} = -30, \quad \frac{dy}{dt} = -20$$

$$\frac{dz}{dt} = ?$$

$$z^2 = x^2 + y^2$$

$$\frac{d}{dt}[z^2] = 2z \frac{dz}{dt} \quad \frac{d}{dt}[x^2 + y^2] = 2x \frac{dx}{dt} + 2y \frac{dy}{dt}$$

$$2z \cdot \frac{dz}{dt} = 2x \frac{dx}{dt} + 2y \frac{dy}{dt}$$

At the moment when $x = 0.4$ and $y = 0.3$:

$$z^2 = (0.4)^2 + (0.3)^2 = 0.16 + 0.09 = 0.25 \Rightarrow z = 0.5$$

$$2 \times 0.5 \cdot \frac{dz}{dt} = 2 \times 0.4 (-30) + 2 \times 0.3 (-20)$$

$$\frac{dz}{dt} = \frac{2 \times 4 \times (-30)}{10} + \frac{2 \times 3 \times (-20)}{10} = -24 - 12 = -36 \text{ mph}$$

2. (Fall 2023, #5) A tank is in the shape of an inverted pyramid with square base (depicted). Its height is 6 m and the length of the side of its base is 3 m. Water is poured into the tank at a rate of $2 \text{ m}^3/\text{min}$. At what rate is the height of the water rising when the height of the water is 4 m?

Hint. The volume of a pyramid whose base is a square of side w and whose height is h is given by the formula $V = w^2h/3$.

$$\frac{dV}{dt} = 2 \quad \frac{dh}{dt} = ?$$

$$V = w^2h/3$$

Since the small pyramid filled with water and the big tank are similar, we have:

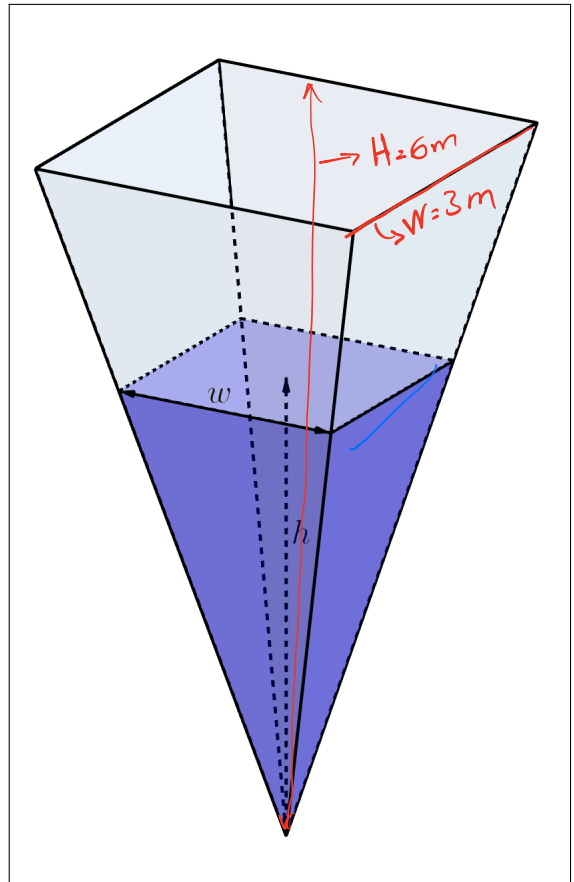
$$\frac{w}{3} = \frac{h}{6} \Rightarrow w = h/2$$

$$\Rightarrow V = \left(\frac{h}{2}\right)^2 h / 3$$

$$\Rightarrow V = \frac{h^3}{12} \Rightarrow \frac{dV}{dt} = \frac{1}{12} \cdot 3 \cdot h^2 \cdot \frac{dh}{dt} = \frac{h^2}{4} \cdot \frac{dh}{dt}$$

We are looking for $\frac{dh}{dt}$ at the moment that $h = 4$.

$$\Rightarrow 2 = \frac{1}{12} \cdot (4)^2 \cdot \frac{dh}{dt} \Rightarrow \frac{dh}{dt} = \frac{2}{4} = \frac{1}{2} \text{ m/min}$$



3. (Spring 2023, #4) The pressure, P (measured in bar), and volume, V (measured in liters), of a sample of helium gas are related via the equation

$$\left(P + \frac{60}{V^2}\right) \cdot (V - 1) = 20$$

At a certain moment in time when the volume of the gas is 2 liters, the pressure of the gas is increasing at a rate of 3 bar per minute. Find the rate of change of volume at that moment in time. Include units in your answer.

$$\frac{dP}{dt} = 3 \text{ b/m} \quad \frac{dV}{dt} = ?$$

$$\left(P + \frac{60}{V^2}\right) (V - 1) = 20$$

$$\Rightarrow \frac{d}{dt} [V-1] = \frac{dV}{dt}, \quad \frac{d}{dt} \left[P + 60 \cdot V^{-2} \right] = \frac{dP}{dt} + 60 \cdot (-2) \cdot V^{-3} \frac{dV}{dt}$$

$$\Rightarrow \frac{d}{dt} \left[\left(P + \frac{60}{V^2}\right) (V-1) \right] = \left(\frac{dP}{dt} - 120 V^{-3} \frac{dV}{dt} \right) (V-1) + \left(\frac{dV}{dt} \right) \left(P + \frac{60}{V^2} \right) = 0$$

At the moment when $V = 2$:

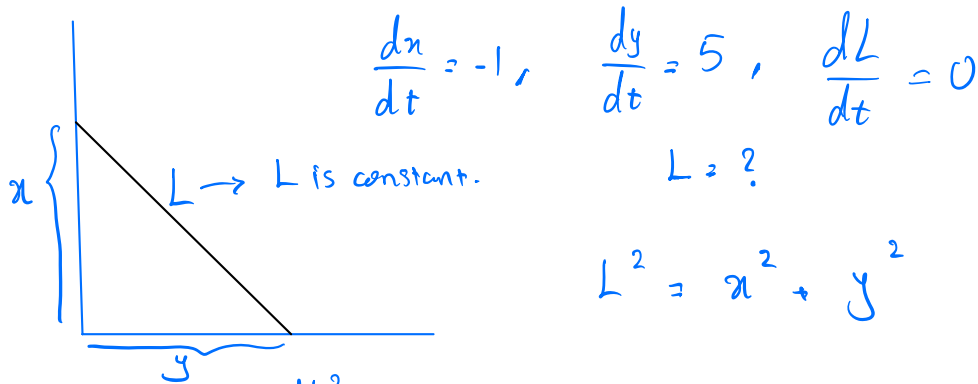
$$\left(P + \frac{60}{2^2}\right) \cdot (2-1) = 20 \Rightarrow P + 15 = 20 \Rightarrow P = 5 \text{ bar}$$

$$\left(3 - \frac{120}{2^3} \frac{dV}{dt}\right) (2-1) + \frac{dV}{dt} \left(5 + \frac{60}{2^2}\right) = 0$$

$$3 - 15 \frac{dV}{dt} + 20 \frac{dV}{dt} = 0 \Rightarrow 5 \frac{dV}{dt} + 3 = 0$$

$$\Rightarrow \frac{dV}{dt} = -\frac{3}{5} \text{ liters per minute.}$$

4. (Fall 2022, #5) A ladder is leaning against a wall and the top is sliding down at a rate of 1 meter per second. At the moment when the bottom of the ladder is 2 meters from the wall, it is sliding away from the wall at a rate of 5 meters per second. How long is the ladder?



Since L^2 is constant $\Rightarrow \frac{dL^2}{dt} = 0$, Hence:

$$0 = 2 \cdot x \cdot \frac{dx}{dt} + 2 \cdot y \cdot \frac{dy}{dt}$$

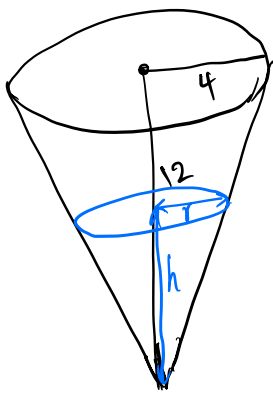
At the moment when $y = 2$:

$$\Rightarrow 2x \cdot (-1) + 2 \times (2) \times 5 = 0$$

$$\Rightarrow 2x = 20 \Rightarrow x = 10$$

$$L^2 = x^2 + y^2 = 10^2 + 2^2 = 104 \Rightarrow L = \sqrt{104} \text{ meters.}$$

5. (Spring 2022, #6) Suppose you have an ice cream cone with radius 4 cm and height 12 cm with some melted boysenberry ice cream in it. The ice cream leaks out of the bottom of the cone at a rate of $8 \text{ cm}^3/\text{min}$. At what rate is the height of the ice cream changing when the height of the ice cream is 2 cm? (Hint: The volume of a cone is given by $V = \frac{\pi}{3} r^2 h$)



$$\frac{dV}{dt} = -8, \quad \frac{dh}{dt} = ?$$

$$V = \frac{\pi}{3} r^2 h$$

Since small and big cones are similar:

$$\frac{r}{4} = \frac{h}{12} \Rightarrow r = \frac{h}{3}$$

$$\Rightarrow V = \frac{\pi}{3} r^2 h = \frac{\pi}{3} \left(\frac{h}{3}\right)^2 h = \frac{\pi h^3}{27}$$

$$\Rightarrow \frac{dV}{dt} = \frac{\pi}{27} \cdot 3 \cdot h^2 \cdot \frac{dh}{dt}$$

At the moment when $h = 2 \text{ cm}$.

$$\begin{aligned} -8 &= \frac{\pi}{\cancel{27} 9} \cdot \cancel{3} \cdot 2^2 \cdot \frac{dh}{dt} \Rightarrow \frac{dh}{dt} = \frac{-\cancel{8} 2}{\frac{\pi}{9} \cdot \cancel{2}^2} \\ &= \frac{-18}{\pi} \text{ cm/min} \end{aligned}$$